



Progress on Defect Predictor

By: Huy Nguyen and Edward Feng



Implementation

- Tested new DefectPredictorTemplate
 - Run new harness.py
- Added new parameters to the calculate risk function
- Implemented & tested all calculate functions as well as the risk score

Revisit Parameters in the Formula

- Recency (Date committed)
- Frequency (How often the files are changed)
- Message keywords
 - Very depending on developer preference so might not work/less weight
- Number of lines modified
 - Lines added + lines deleted
- How many lines of code (loc) in the file
- Cyclomatic Complexity **NEW**

Revisit Pydriller

- Looked into Pydriller docs
- Has pretty much everything we need
 - added_lines & deleted_lines for calculate_modified_lines()
 - nloc for calculate_lines_of_code()
 - complexity for calculate_cyclomatic_complexity()

ModifiedFile

You can get a list of modified files as well as their diffs and current source code from each commit. All *Modifications* can be obtained by iterating over the ModifiedFile object. Each modification object references a modified file and has the following fields:

- **old_path**: old path of the file (can be `None` if the file is added)
- **new_path**: new path of the file (can be `None` if the file is deleted)
- **filename**: return only the filename (e.g., given a path-like-string such as `"/Users/dspadini/pydriller/myfile.py"` returns `"myfile.py"`)
- **change_type**: type of the change: can be Added, Deleted, Modified, or Renamed. If you use `change_type.name` you get `ADD`, `DELETE`, `MODIFY`, `RENAME`.
- **diff**: diff of the file as Git presents it (e.g., starting with `@@ xx,xx @@`).
- **diff_parsed**: diff parsed in a dictionary containing the added and deleted lines. The dictionary has 2 keys: "added" and "deleted", each containing a list of Tuple (int, str) corresponding to (number of line in the file, actual line).
- **added_lines**: number of lines added
- **deleted_lines**: number of lines removed
- **source_code**: source code of the file (can be `None` if the file is deleted or only renamed)
- **source_code_before**: source code of the file before the change (can be `None` if the file is added or only renamed)
- **methods**: list of methods of the file. The list might be empty if the programming language is not supported or if the file is not a source code file. These are the methods **after** the change.
- **methods_before**: list of methods of the file **before** the change (e.g., before the commit.)
- **changed_methods**: subset of `methods` containing **only** the changed methods.
- **nloc**: Lines Of Code (LOC) of the file
- **complexity**: Cyclomatic Complexity of the file
- **token_count**: Number of Tokens of the file

Testing

- Ran the new sample repo
- Tested all calculate functions
 -
- Tested calculate_risk() without recency for now
 - Results are too imbalanced
- For demo, we will be running the old sample repo because it is faster

Calculating recency scores...

```
Recency Scores (File Name -> Recency Value):
modules/ui.py -> 2025-02-03 20:33:44+09:00
modules/metadata.py -> 2025-01-06 00:27:45+08:00
modules/processors/frame/face_swapper.py -> 2025-03-10 23:41:28+02:00
modules/globals.py -> 2025-02-03 20:33:44+09:00
modules/processors/frame/core.py -> 2024-07-07 14:33:32+02:00
modules/core.py -> 2024-12-24 09:51:32+03:30
modules/predictor.py -> 2024-08-30 22:02:23+02:00
modules/capturer.py -> 2024-09-06 20:59:35+02:00
modules/processors/frame/face_enhancer.py -> 2024-12-23 06:29:36+00:00
modules/face_analyser.py -> 2025-02-03 20:33:44+09:00
```

Calculating frequency scores...

```
Frequency Scores (File Name -> Frequency Value):
modules/ui.py -> 46
modules/metadata.py -> 9
modules/processors/frame/face_swapper.py -> 8
modules/globals.py -> 10
modules/processors/frame/core.py -> 1
modules/core.py -> 5
modules/predictor.py -> 3
modules/capturer.py -> 2
modules/processors/frame/face_enhancer.py -> 6
modules/face_analyser.py -> 1
```

Calculating modified scores...

```
Modified Scores (File Name -> Modified Value):
modules/ui.py -> 4354
modules/metadata.py -> 22
modules/processors/frame/face_swapper.py -> 45
modules/globals.py -> 47
modules/processors/frame/core.py -> 3
modules/core.py -> 299
modules/predictor.py -> 16
modules/capturer.py -> 42
modules/processors/frame/face_enhancer.py -> 94
modules/face_analyser.py -> 24
```

Calculating LOC (lines of code) scores...

```
LOC (lines of code) Scores (File Name -> LOC Value):
modules/ui.py -> 220
modules/metadata.py -> 3
modules/processors/frame/face_swapper.py -> 67
modules/globals.py -> 27
modules/processors/frame/core.py -> 62
modules/core.py -> 259
modules/predictor.py -> 19
modules/capturer.py -> 20
modules/processors/frame/face_enhancer.py -> 75
modules/face_analyser.py -> 147
```

Calculating cyclomatic complexity scores...

```
Complexity Scores (File Name -> Complexity Value):
modules/ui.py -> 40
modules/metadata.py -> 0
modules/processors/frame/face_swapper.py -> 20
modules/globals.py -> 0
modules/processors/frame/core.py -> 17
modules/core.py -> 68
modules/predictor.py -> 5
modules/capturer.py -> 6
modules/processors/frame/face_enhancer.py -> 17
modules/face_analyser.py -> 48
```

Calculating risk scores...

```
Risk Scores (File Name -> Risk Value):
modules/ui.py -> 912.90
modules/metadata.py -> 7.85
modules/processors/frame/face_swapper.py -> 20.50
modules/globals.py -> 15.60
modules/processors/frame/core.py -> 8.85
modules/core.py -> 94.25
modules/predictor.py -> 6.65
modules/capturer.py -> 11.70
modules/processors/frame/face_enhancer.py -> 30.10
modules/face_analyser.py -> 24.65
```

```
risk_score = ( .35 * f ) + ( .2 * m ) + ( .1 * c ) + ( .1 * loc )
```

What's next?

- Entering Sprint 4
 - Collect feedback and apply fixes
 - Fix calculate_recency
 - So that it returns numbers (to_date - commit_date) instead of Date Strings
 - Running into issues due to timezone
 - Normalize values
 - Redistribute Weight
 - Print the top X results
 - Update README

```
Calculating recency scores...
```

```
2025-03-03 00:00:00
```

```
Recency Scores (File Name -> Recency Value):
```

```
buggy_script.py -> 2025-02-28 19:53:53-05:00
```

```
word_frequency.py -> 2025-03-02 18:05:42-05:00
```